

■ Function

`Function[body]` or `body&` is a pure function. The formal parameters are `#` (or `#1`), `#2`, etc.

`Function[x, body]` is a pure function with a single formal parameter `x`.

`Function[{x1, x2, ...}, body]` is a pure function with a list of formal parameters.

Example: `(# + 1)&[x] → 1 + x`. ■ `Map[(# + 1)&, {x, y, z}] → {1 + x, 1 + y, 1 + z}`. ■ When `Function[body]` or `body&` is applied to a set of arguments, `#` (or `#1`) is replaced by the first argument, `#2` by the second, and so on. ■ If there are more arguments supplied than `#i` in the function, the remaining arguments are ignored. ■ `Function` has attribute `HoldAll`. The function body is evaluated only after the formal parameters have been replaced by arguments.

■ The named formal parameters `xi` in `Function[{x1, ...}, body]` are treated as local. ■ `##` stands for the sequence of all arguments supplied. ■ `##n` stands for arguments from number `n` on.

■ `f[##, ##2]&[x, y, z] → f[x, y, z, y, z]`. ■ `Function` is analogous to λ in LISP or formal logic. ■ Formal parameters inside nested `Function` objects are not modified when the outer objects are applied. ■ See page 179. ■ See also: `Apply`.